

MATH I - Week 1 NOTES

Part 3, on Number Systems

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I. Distinguish between numeral systems, number sets and number systems

The terms **numeral systems**, **number sets**, and **number systems** refer to related but different ideas.

A **numeral system** is a method for writing or representing numbers using particular symbols and rules. Examples include the decimal system base 10, binary system base 2, hexadecimal system base 16, and Roman numerals. Thus, the same number can have different representations:

$$13_{10} = 1101_2 = D_{16} = XIII.$$

A **number set** is a collection of numbers having specified properties. The standard number sets include the natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, and complex numbers $\mathbb{C}$, with

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The term **number system** is broader. It can refer to a number set together with the operations and rules defined on it—for example, the real number system includes $\mathbb{R}$, arithmetic operations, ordering, and algebraic properties.

However, “number system” is sometimes also used informally to mean a numeral system. To avoid ambiguity, it is best to use **numeral system** for binary, decimal, and hexadecimal representations, and **number sets** or **number domains** for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$.

Terminology (summarizing table):

Concept	Preferred term	Examples
Mathematical domains	Number systems or number sets	Natural, integer, rational, real and complex numbers
Methods of representation	Numeral systems or number-representation systems	Binary, decimal, octal and hexadecimal
Positional base	Base-<i>b</i> numeral system	Base 2, base 10 and base 16
Concrete written symbols	Numerals	17, 10001 ₂ , 11 ₁₆
Abstract mathematical quantity	Number	Seventeen, independently of how it is written

For example, the same number may have several numeral representations:

$$17_{10} = 10001_2 = 21_8 = 11_{16}.$$

Here, the abstract **number** is unchanged; only its **numeral** changes.

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Understanding Number Systems

Learning Objectives

After completing Week 1 of the Math I course, you should be able to

- understand why different kinds of numbers exist;
 - distinguish between natural, integer, rational, irrational, and real numbers;
 - understand why the real numbers form the foundation of calculus;
 - understand why engineers care about real numbers;
 - understand complex numbers.
-

1. Why do we need numbers?

Suppose an engineer asks

- How long is this bridge?
- What is the voltage?
- How much energy is stored?
- How fast is the aircraft?

All these questions require numbers.

Numbers allow us to

- measure,
- compare,
- calculate,
- predict,
- optimize.

Engineering is impossible without numbers!

2. The evolution of numbers

Human civilization gradually invented new kinds of numbers whenever old ones became insufficient.

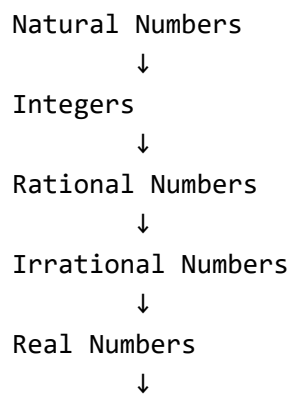
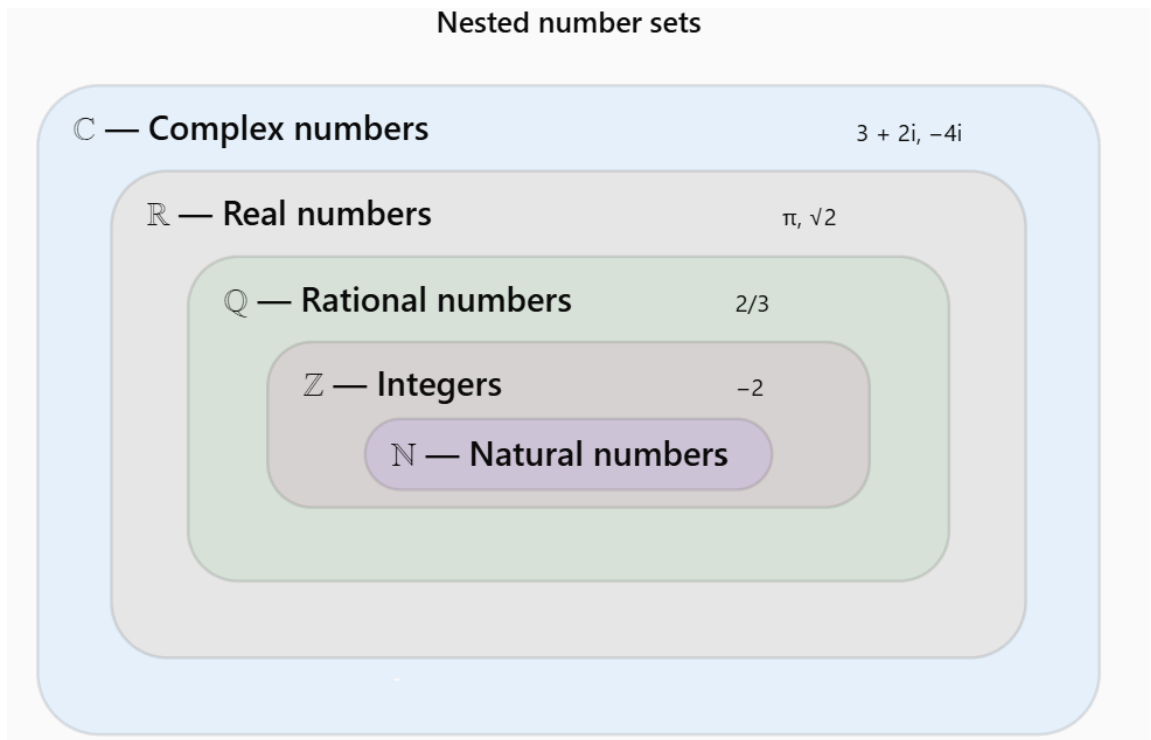


Table and visualization (Venn diagrams):

Set Name	Set Symbol	Set Description
Natural Numbers	$\mathbb{N}$	The counting numbers
Whole Numbers		The counting numbers and 0
Integers	$\mathbb{Z}$	The natural numbers, their negatives, and 0
Rational Numbers	$\mathbb{Q}$	Fractions of integers
Irrational Numbers	$\mathbb{P}$	Numbers that cannot be written as a fraction of integers
Real Numbers	$\mathbb{R}$	The union of the rational and irrational numbers, all possible physical lengths, and their negatives

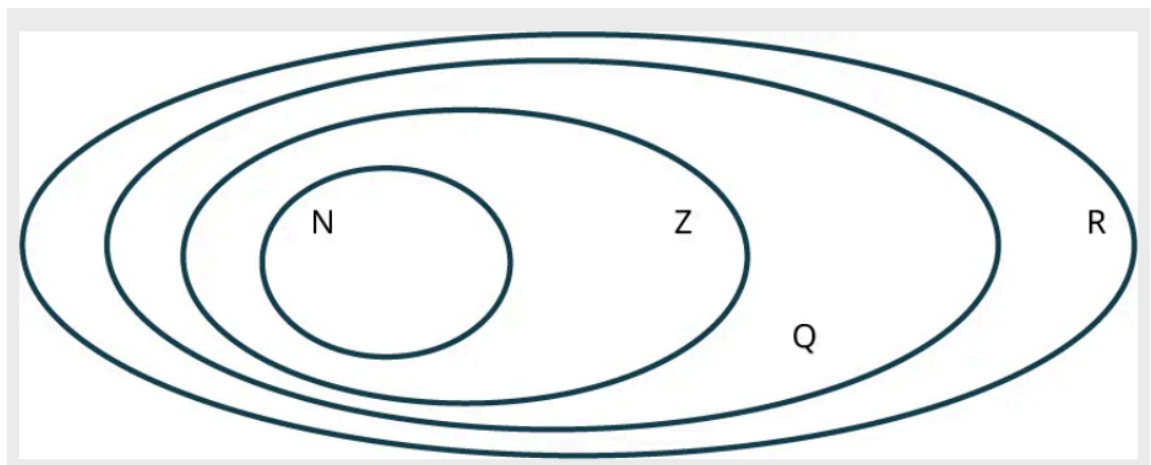
This nested “Venn-type” diagram shows both the containment relations and examples belonging specifically to each region.

Nested number sets



The essential relationship is

$$\boxed{N \subseteq Z \subseteq Q \subseteq R \subseteq C}.$$



A note about notation in set difference:

For two sets A and B ,

$$A \setminus B$$

means “the elements belonging to A but not to B ”:

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

As another example,

$$\mathbb{R} \setminus \mathbb{Q},$$

is the set of irrational numbers.

Can we write?

$$\mathbb{R} - \mathbb{Q},$$

We could and many readers will understand it as the set difference. However,

$$\boxed{\mathbb{R} \setminus \mathbb{Q}}$$

is the clearer and more standard notation, particularly because the minus sign may be confused with ordinary numerical subtraction.

Thus, prefer:

$$\mathbb{R} \setminus \mathbb{Q}$$

which denotes the irrational numbers:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

The notation $\mathbb{R} - \mathbb{Q}$ is acceptable if set subtraction has been explicitly defined, but $\setminus$ is preferable in textbooks and formal mathematical writing.

3. Natural Numbers

The first numbers humans used were

1, 2, 3, 4, 5, ...

They answer

"How many?"

Examples

- 5 sensors
- 20 bolts
- 8 robots

Natural numbers are used for counting.

Notation: $\mathbb{N}$

4. Integers

Natural numbers cannot answer

$5 - 8$?

The answer is -3 .

Negative numbers were invented.

Integers are

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Notation: $\mathbb{Z}$

Examples

- temperature: -15°C
 - depth: -50m
 - electric charge: positive or negative.
-

5. Rational Numbers

Integers cannot solve

$$1 \div 2 (= \frac{1}{2})$$

We need fractions.

Examples

$$\frac{1}{2}, \frac{3}{5}, -\frac{7}{4}$$

Notation: $\mathbb{Q}$

Every rational number can be written as

$$\frac{a}{b}$$

where

- a is an integer
- $b \neq 0$

Examples

$$0.25 = \frac{1}{4}$$

$$0.333333 \dots = \frac{1}{3}$$

Rational numbers have

- terminating decimals

or

- repeating decimals.

6. A surprising discovery

The ancient Greeks believed everything could be expressed as a fraction. Then they studied a square. Suppose the side length equals 1.

Using Pythagoras, the diagonal equals $\sqrt{2}$.

Can $\sqrt{2}$ be written as $\frac{a}{b}$?

The answer is No. This was shocking.

7. Irrational Numbers

Irrational numbers cannot be written as fractions.

Examples: $\sqrt{2}$, $\sqrt{3}$, π , e

Their decimal expansions never end and never repeat.

Example: $\pi = 3.1415926535\dots$

8. Real Numbers

Real numbers combine

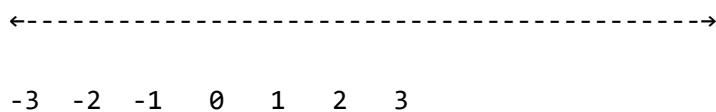
- rational numbers

and

- irrational numbers.

Notation: $\mathbb{R}$

Everything on the continuous number line is a real number.



Every point corresponds to one real number.

Real numbers represent

- length
- time
- mass
- voltage
- pressure
- temperature
- velocity

Almost every engineering measurement is a real number.

9. Why are real numbers important?

Calculus studies continuous change. Suppose

- temperature changes continuously.
- Position changes continuously.
- Voltage changes continuously.
- Current changes continuously.

These quantities require continuous numbers.

The real numbers make continuity possible!

10. Density of the real numbers

Between any two real numbers there are infinitely many others.

Example between 1 and 2 we find 1.5, between 1 and 1.5 we find 1.25, between 1 and 1.25 we find 1.125.

This process never ends. ***Real numbers are infinitely dense.***

11. The Number Line

Every real number corresponds to exactly one point.

Likewise, every point corresponds to one real number. This correspondence is fundamental.

Calculus transforms geometry into algebra.

12. The properties of real numbers

Real numbers satisfy important rules.

Commutative Law

Addition

$$a + b = b + a$$

Multiplication

$$ab = ba$$

Associative Law

$$(a + b) + c = a + (b + c)$$

$$(ab)c = a(bc)$$

Distributive Law

$$a(b + c) = ab + ac$$

Identity Elements

Addition

$$a + 0 = a$$

Multiplication

$$a \cdot 1 = a$$

Inverse Elements

Addition

$$a + (-a) = 0$$

Multiplication

$$a \cdot \frac{1}{a} = 1$$

for

$$a \neq 0.$$

13. Ordering real numbers

Real numbers can be compared.

Examples

$$2 < 5$$

$$-3 < 1$$

Ordering allows us to define

- maximum
- minimum
- intervals

which become essential in calculus.

14. Intervals

Examples

Closed interval

$$[a, b]$$

contains both endpoints.

Open interval

(a, b)

does not contain the endpoints.

Half-open intervals

$[a, b)$

$(a, b]$

These appear throughout calculus.

15. An important point: Real numbers and computers

Here comes the surprise. Computers do **not** store real numbers. Instead, they store approximations.

Example (in Python)

```
0.1
```

is actually stored approximately.

Therefore

```
0.1 + 0.2
```

returns

```
0.30000000000000004
```

This is not a bug. It is a consequence of binary floating-point arithmetic. But this is something to discuss later in your studies.

Why engineers must know this

Suppose an aircraft simulation performs 100 million calculations. Tiny rounding errors accumulate. Understanding numerical precision becomes essential. Computational engineering begins here.

An interesting point: Numbers in AI

Machine learning models store millions or billions of numbers. AI is essentially large-scale numerical computation.

16. Numbers in Engineering

Examples

Materials Engineering

- elastic modulus
- grain size
- density

Electrical Engineering

- voltage
- current
- impedance

Power Engineering

- frequency
- power
- phase angle

Industrial Engineering

- costs
- probabilities
- waiting times

Everything is expressed with numbers.

17. Summary

The hierarchy of number systems is

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Real numbers

- describe continuous quantities;
- form the foundation of calculus;
- model nearly all engineering measurements;
- are approximated by computers using floating-point arithmetic (to be discussed in another course).

Understanding the real number system is therefore the first step toward mathematical analysis, scientific computing, artificial intelligence, and modern engineering.

Key Takeaways

- **Natural numbers** count objects.
- **Integers** include positive, negative numbers and zero.
- **Rational numbers** are fractions with terminating or repeating decimal expansions.
- **Irrational numbers** cannot be expressed as fractions and have non-repeating decimal expansions.
- **Real numbers** include both rational and irrational numbers and represent continuous quantities.
- **Calculus is built on the real number system.**
- **Computers approximate real numbers**, making numerical precision an essential topic in engineering and AI.

- **Modern engineering and AI are fundamentally numerical disciplines**, relying on accurate representation and computation of real numbers.

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In-Class activity

Try to solve the following problems. Keep your solutions and check them with the answers. The in-class activity is to facilitate your learning; it will not be graded.

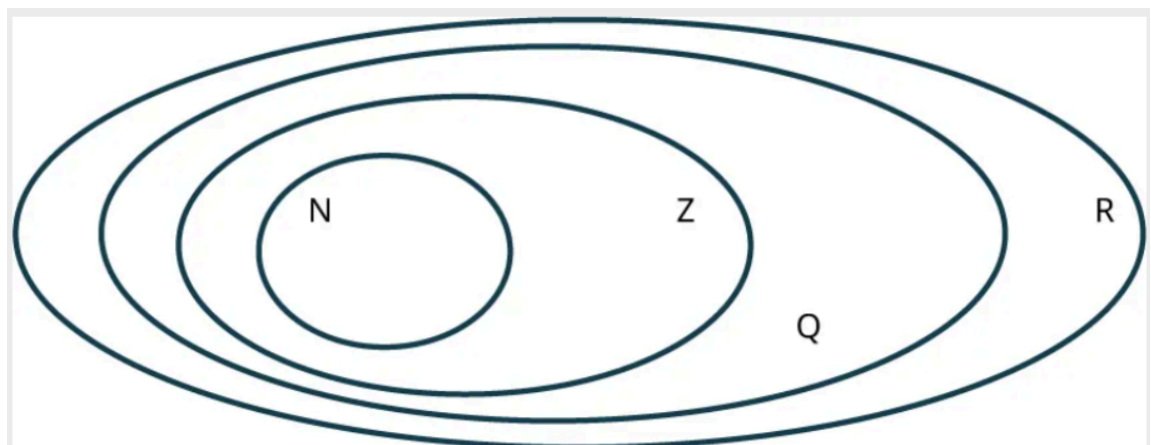
Exercise 1: Set diagram

Draw a nested-set diagram representing

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Place each of the following numbers in the correct region:

$$2026, \quad -9, \quad \frac{2}{3}, \quad \sqrt{10}, \quad \pi, \quad -5i, \quad -2 + 2i.$$



Exercise 2: Rational or irrational?

Determine whether each number is rational or irrational. Justify your answer.

a. $\sqrt{49}$,

b. $\sqrt{18}$,

c. $3 + \sqrt{2}$,

d. $2\sqrt{5}$,

e. $\sqrt{2} + \sqrt{8}$,

f. $(\sqrt{5})^2$

Complex Numbers

1. Why complex numbers are needed

Some algebraic equations have no solutions in the real numbers. For example,

$$x^2 + 1 = 0$$

requires $x^2 = -1$, but no real number has a negative square. We therefore introduce the **imaginary unit**

$$i = \sqrt{-1}, \quad i^2 = -1.$$

A **complex number** has the form

$$z = a + bi,$$

where $a, b \in \mathbb{R}$. The number a is the **real part**, and b is the **imaginary part**:

$$\operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b.$$

For example, if $z = 4 - 3i$, then

$$\operatorname{Re}(z) = 4, \quad \operatorname{Im}(z) = -3.$$

The set of complex numbers is denoted by

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

Every real number is also complex because

$$a = a + 0i.$$

Thus,

$$\mathbb{R} \subseteq \mathbb{C}.$$

2. Equality of complex numbers

Two complex numbers are equal if their real parts are equal and their imaginary parts are equal:

$$a + bi = c + di$$

if and only if

$$a = c \quad \text{and} \quad b = d.$$

For example, if

$$(2x - 1) + (y + 3)i = 5 - 2i,$$

then

$$2x - 1 = 5, \quad y + 3 = -2.$$

Therefore,

$$x = 3, \quad y = -5.$$

3. Powers of i

The powers of i repeat every four exponents:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i,$$

and

$$i^4 = 1.$$

The cycle then repeats:

$$i^{n+4} = i^n.$$

To calculate a large power of i , divide the exponent by 4 and use the remainder.

For example,

$$37 = 4(9) + 1,$$

so

$$i^{37} = i^1 = i.$$

Similarly,

$$i^{202} = i^{4(50)+2} = i^2 = -1.$$

4. Algebraic operations

Let

$$z_1 = a + bi, \quad z_2 = c + di.$$

4.1 Addition

Add the real parts and imaginary parts separately:

$$z_1 + z_2 = (a + c) + (b + d)i.$$

Example:

$$(3 + 2i) + (5 - 7i) = 8 - 5i.$$

4.2 Subtraction

$$z_1 - z_2 = (a - c) + (b - d)i.$$

Example:

$$(3 + 2i) - (5 - 7i) = -2 + 9i.$$

4.3 Multiplication

Use the distributive law and $i^2 = -1$:

$$(a + bi)(c + di) = ac + adi + bci + bdi^2.$$

Therefore,

$$\boxed{(a + bi)(c + di) = (ac - bd) + (ad + bc)i}.$$

Example:

$$(2 + 3i)(4 - i) = 8 - 2i + 12i - 3i^2 = 11 + 10i.$$

4.4 Division

To divide complex numbers, multiply the numerator and denominator by the conjugate of the denominator.

For example,

$$\frac{3 + 2i}{1 - i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)}.$$

The denominator is

$$(1 - i)(1 + i) = 1 - i^2 = 2.$$

The numerator is

$$(3 + 2i)(1 + i) = 3 + 3i + 2i + 2i^2 = 1 + 5i.$$

Hence,

$$\frac{3 + 2i}{1 - i} = \boxed{\frac{1}{2} + \frac{5}{2}i}.$$

5. Complex conjugate

The **complex conjugate** of

$$z = a + bi$$

is

$$\boxed{\bar{z} = a - bi}.$$

Geometrically, conjugation reflects a complex number across the real axis.

Important properties include

$$z + \bar{z} = 2 \operatorname{Re}(z),$$

$$z - \bar{z} = 2i \operatorname{Im}(z),$$

and

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2.$$

Also,

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2,$$

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2.$$

6. The complex plane

A complex number

$$z = a + bi$$

can be represented by the point (a, b) in the **complex plane**, also called the **Argand plane**.

- The horizontal axis is the real axis.
- The vertical axis is the imaginary axis.

Thus,

$$3 + 2i \longleftrightarrow (3, 2).$$

Addition of complex numbers corresponds geometrically to vector addition.

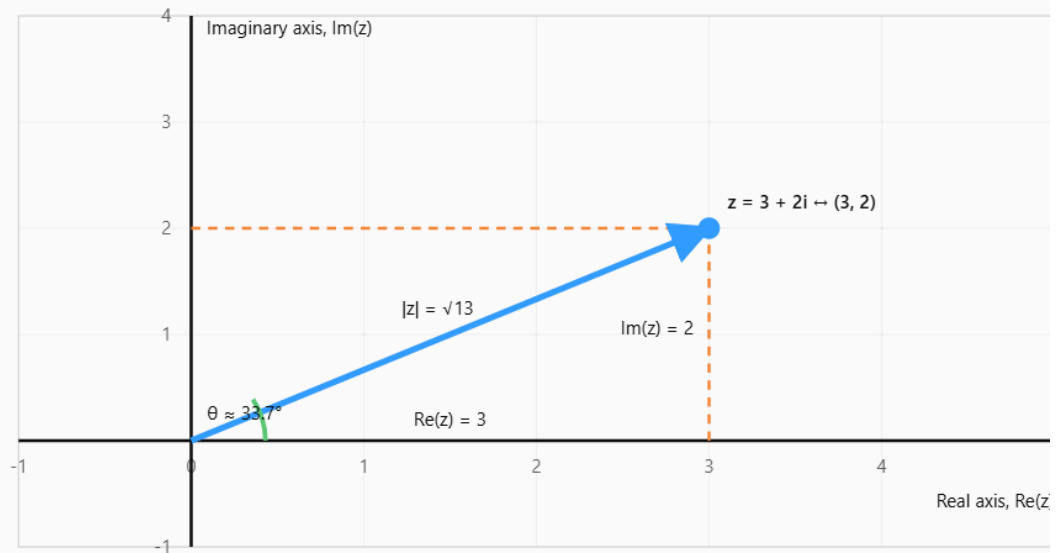
Example:

For a complex number

$$z = a + bi,$$

the horizontal coordinate represents the real part a , while the vertical coordinate represents the imaginary part b . The diagram uses $z = 3 + 2i$.

Complex number on the complex plane: $z = 3 + 2i$



Thus,

$$\operatorname{Re}(z) = 3, \quad \operatorname{Im}(z) = 2.$$

The distance from the origin is the **modulus**:

$$|z| = \sqrt{3^2 + 2^2} = \sqrt{13},$$

and the angle with the positive real axis is the **argument**:

$$\arg(z) = \tan^{-1}\left(\frac{2}{3}\right) \approx 33.7^\circ.$$

...Who invented complex numbers?

Complex numbers were not invented by a single person. They developed gradually:

- *Girolamo Cardano* encountered square roots of negative numbers while solving cubic equations in *Ars Magna* (1545), although he regarded them as puzzling and of little practical value.
- *Rafael Bombelli* developed systematic rules for calculating with them in *L'Algebra* (1572). For this reason, Bombelli is often regarded as the most important early founder of complex-number arithmetic.
- *René Descartes* introduced the term *imaginary numbers* in 1637.
- *Abraham de Moivre* connected them with trigonometry through de Moivre's formula.
- *Leonhard Euler* established the notation $i = \sqrt{-1}$ and the fundamental connection

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

- *Caspar Wessel* presented a geometric plane interpretation in 1799. *Jean-Robert Argand* independently developed and popularized it, giving rise to the name *Argand diagram*.
- *Carl Friedrich Gauss* helped establish complex numbers as legitimate mathematical objects in the nineteenth century.

7. Modulus of a complex number

The **modulus** or absolute value of $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$

It represents the distance between the point z and the origin.

For example,

$$|3 - 4i| = \sqrt{3^2 + (-4)^2} = 5.$$

Because

$$z\bar{z} = a^2 + b^2,$$

we obtain

$$z\bar{z} = |z|^2.$$

Consequently, if $z \neq 0$,

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Important modulus properties are

$$|z_1 z_2| = |z_1| |z_2|,$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad z_2 \neq 0,$$

and the triangle inequality

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

8. Argument of a complex number

For a nonzero complex number $z = a + bi$, the **argument** of z is an angle θ measured from the positive real axis to the vector representing z .

We write

$$\theta = \arg z.$$

Since adding a full rotation does not change the direction,

$$\arg z = \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

A selected value, usually in

$$(-\pi, \pi],$$

is called the **principal argument** and is denoted by

$$\text{Arg}(z).$$

Although

$$\tan \theta = \frac{b}{a},$$

the quadrant containing (a, b) must be considered.

For example, for

$$z = -1 + i,$$

the point lies in the second quadrant. Therefore,

$$|z| = \sqrt{2}, \quad \text{Arg}(z) = \frac{3\pi}{4}.$$

The number 0 has modulus 0, but its argument is undefined.

9. Algebraic, trigonometric and exponential forms

9.1 Algebraic form

The usual form is

$$\boxed{z = a + bi}.$$

9.2 Trigonometric form

If $r = |z|$ and $\theta = \arg z$, then

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Therefore,

$$\boxed{z = r(\cos \theta + i \sin \theta)}.$$

This is the **trigonometric** or **polar form**.

For example, consider

$$z = 1 + \sqrt{3}i.$$

Its modulus is

$$r = \sqrt{1^2 + (\sqrt{3})^2} = 2.$$

Since the point is in the first quadrant,

$$\theta = \frac{\pi}{3}.$$

Thus,

$$\boxed{1 + \sqrt{3}i = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}.$$

9.3 Euler's formula

Euler's formula states that

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

It gives the famous identity

$$e^{i\pi} + 1 = 0.$$

9.4 Exponential form

Using Euler's formula, the trigonometric form becomes

$$z = re^{i\theta}.$$

Thus,

$$1 + \sqrt{3}i = 2e^{i\pi/3}.$$

Because the argument is not unique, all equivalent exponential forms are

$$z = re^{i(\theta+2k\pi)}, \quad k \in \mathbb{Z}.$$

10. Multiplication and division in polar form

Let

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2}.$$

Then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}.$$

Therefore, multiplication:

- multiplies the moduli;
- adds the arguments.

Similarly,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}, \quad z_2 \neq 0.$$

Division:

- divides the moduli;
- subtracts the arguments.

Geometrically, multiplication by $re^{i\theta}$ scales a complex number by r and rotates it through the angle θ .

11. de Moivre's formula

For every integer n ,

$$\boxed{(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)}.$$

Equivalently, if

$$z = r(\cos \theta + i \sin \theta),$$

then

$$\boxed{z^n = r^n (\cos(n\theta) + i \sin(n\theta))}.$$

In exponential form,

$$(re^{i\theta})^n = r^n e^{in\theta}.$$

Example: Calculate $(1 + i)^8$

First write $1 + i$ in polar form:

$$|1 + i| = \sqrt{2}, \quad \arg(1 + i) = \frac{\pi}{4}.$$

Therefore,

$$1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

By de Moivre's formula,

$$(1 + i)^8 = (\sqrt{2})^8 \left(\cos \frac{8\pi}{4} + i \sin \frac{8\pi}{4} \right).$$

Since

$$(\sqrt{2})^8 = 16, \quad \cos(2\pi) = 1, \quad \sin(2\pi) = 0,$$

we get

$$\boxed{(1 + i)^8 = 16}.$$

Deriving trigonometric identities

For $n = 2$,

$$(\cos \theta + i \sin \theta)^2 = \cos(2\theta) + i \sin(2\theta).$$

Expanding the left side gives

$$\cos^2 \theta - \sin^2 \theta + 2i \sin \theta \cos \theta.$$

Comparing real and imaginary parts produces

$$\boxed{\cos(2\theta) = \cos^2 \theta - \sin^2 \theta},$$

$$\boxed{\sin(2\theta) = 2 \sin \theta \cos \theta}.$$

12. Roots of complex numbers

Suppose we want to solve

$$z^n = w,$$

where

$$w = re^{i\theta}.$$

Write

$$z = \rho e^{i\phi}.$$

Then

$$z^n = \rho^n e^{in\phi}.$$

Therefore,

$$\rho^n = r \implies \rho = r^{1/n},$$

and

$$n\phi = \theta + 2k\pi.$$

Hence the n distinct roots are

$$\boxed{z_k = r^{1/n} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right]},$$

where

$$k = 0, 1, \dots, n-1.$$

The roots have equal modulus and are equally spaced around a circle.

Example: Cube roots of 8

Write

$$8 = 8e^{i(2k\pi)}.$$

The cube roots have modulus

$$8^{1/3} = 2$$

and arguments

$$\frac{2k\pi}{3}, \quad k = 0, 1, 2.$$

Thus,

$$z_0 = 2,$$

$$z_1 = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = -1 + \sqrt{3}i,$$

$$z_2 = 2 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) = -1 - \sqrt{3}i.$$

Therefore,

$$z = 2, -1 + \sqrt{3}i, -1 - \sqrt{3}i.$$

13. Roots of unity

The n -th roots of unity solve

$$z^n = 1.$$

Since

$$1 = e^{i2k\pi},$$

the roots are

$$z_k = e^{2\pi i k/n} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n},$$

where $k = 0, \dots, n-1$.

For example, the fourth roots of unity are

$$1, i, -1, -i.$$

Geometrically, the n -th roots of unity form the vertices of a regular n -gon on the unit circle.

14. Solving quadratic equations

The quadratic formula remains valid over $\mathbb{C}$:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

For example,

$$z^2 - 4z + 13 = 0$$

has discriminant

$$\Delta = (-4)^2 - 4(1)(13) = -36.$$

Since

$$\sqrt{-36} = 6i,$$

we obtain

$$z = \frac{4 \pm 6i}{2}.$$

Therefore,

$$\boxed{z = 2 + 3i \quad \text{or} \quad z = 2 - 3i}.$$

If a polynomial has real coefficients, its non-real roots occur in conjugate pairs.

15. Fundamental Theorem of Algebra

The **Fundamental Theorem of Algebra** states:

Every nonconstant polynomial with complex coefficients has at least one complex root.

Consequently, a polynomial of degree n has exactly n complex roots when roots are counted according to their multiplicities.

For example,

$$p(z) = z^3 - z^2 + z - 1$$

can be factored as

$$p(z) = (z - 1)(z^2 + 1) = (z - 1)(z - i)(z + i).$$

Its three complex roots are

$$\boxed{1, i, -i}.$$

In-Class activity

Try to solve the following problems. Keep your solutions and check them with the answers provided later (and shown below). The in-class activity is to facilitate your learning; it will not be graded.

Exercise 1: Basic complex arithmetic

Given

$$z_1 = 2 + 3i, \quad z_2 = 1 - 5i,$$

calculate:

$$z_1 + z_2, \quad z_1 - z_2, \quad z_1 z_2, \quad \frac{z_1}{z_2}.$$

Write every answer in the form $a + bi$.

Exercise 2: Explain why

$$1 + i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2e^{i\frac{\pi}{3}}$$

TO THE READER:

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